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Paleomagnetic study of the Eocene Quxu pluton of the Gangdese Belt: Crustal deformation along the Indus–Zangbo suture zone in southern Tibet

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ABSTRACT

Forty-five samples have been collected at nine sites on the 42 5 Ma Quxu pluton (90°50′E, 29°20′N) in the Gangdese batholith Westerly declination ($D = -48°$ and $-83°$) is observed in primary magnetizations from two sites about 25 km from the Indus–Zangbo suture zone after thermal demagnetization This direction is consistent with the westerly paleomagnetic directions of the crustal blocks in other areas along the Indus–Zangbo suture zone The Quxu pluton of the Gangdese Belt was rotated in a "domino style" deformation process as a part of a long (840 km) and narrow (less than 100 km) deformed zone between the India–Eurasia continents associated with the collision of India since 42 5 Ma The pluton, between 11 km and 14 km from the suture acquired the secondary magnetization ($D = -28°$ and $-39°$) during a cataclastic metamorphic process at sometime during the 'domino style' deformation. The primary magnetization was completely destroyed in the pluton within 11 km of the suture during slow cooling at the uplift stage and was replaced by thermoviscous remanent magnetization parallel to the present axial dipole field

1. Introduction

The collision of the Indian and Asian continents provides information on the deformation of continental lithosphere. The Indian continent collided with the Asian continent at about 45 Ma [1–6]. The Indus–Zangbo suture zone is the site of the collision between these two continents. Significant crustal deformation has been observed in the narrow area along the Indus–Zangbo suture zone. Paleomagnetic studies showed that five areas between 80°E and 89°E along the Indus–Zangbo suture zone were rotated counter-clockwise through about 60° [7,8]. Block rotation along the

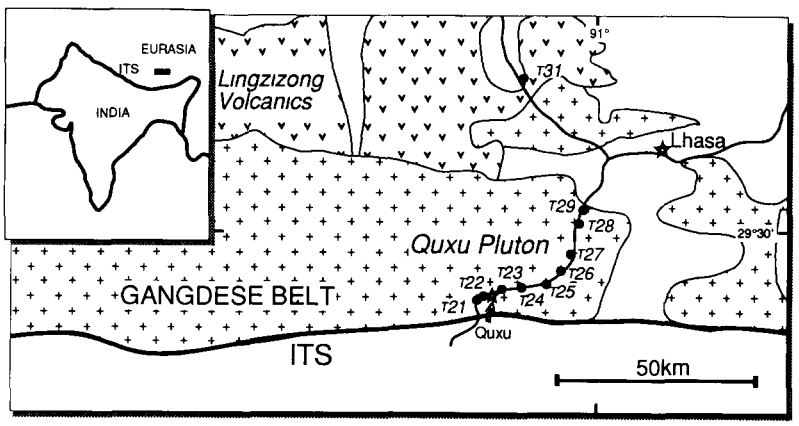


Fig. 1 Simplified geological map of the Southern Lhasa Terrane in the vicinity of Lhasa, and localities of sampling sites of the Quxu pluton (T21–T29) and the Lingzizong volcanics (T31) (Table 1) Base map from [16]

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Indus–Zangbo suture zone is one of prominent examples of deformation phenomena of the continental lithosphere in southern Tibet.

North of the Indus–Zangbo suture lies the Gangdese batholith (Fig. 1), which constitutes a major east–west trending plutonic belt in southern Tibet The Gangdese Belt was an active continental arc at the southern edge of Asia until the collision of India. The deformation phenomena which occurred in front of the colliding India is probably recorded in this belt.

We studied paleomagnetism of the Quxu plutons in the Gangdese Belt in order to detect evidence of deformation along the Indus–Zangbo suture zone. Because this pluton is one of the most studied rock bodies in Tibet [9,10] it provides a magnificent opportunity for studying the timing of deformation and extent of the deformation area along the suture zone.

2. Geology and sampling

The Gangdese batholith runs parallel to the Indus–Zangbo suture zone over a distance of about 2600 km. On the basis of chemical–mineralogical typology the genesis of the batholith is related to oceanic subduction [9,11]. Published age data demonstrate that its crystallization ages range from 100 Ma to 40 Ma [5,6,10,12-14]. For the eastern part of the Gangdese Belt (the Quxu pluton), geochronological studies of $^{39}Ar/^{40}Ar$ and U-Pb gave numerous Cenozoic ages close to 42.5 Ma [10]. Many workers have argued that the eastern part of the Gangdese batholith was emplaced a short time before and during the beginning of the India–Asia collision associated with subduction of the Tethys seafloor [11,15].

The Quxu pluton of the Gangdese batholith intrudes Cretaceous sediments (the Takena formation) in the Lhasa area [16,17] (Fig. 1). The magmatism which was responsible for the origin of the Quxu pluton also produced a thick volcanic sequence of andesite lavas and pyroclastic flows (the Lingzizong formation), which has been dated at 48.5 Ma [18]. This volcanic sequence unconformably overlies the folded Takena formation.

In 1986 we sampled granitic rocks of the Quxu pluton at nine sites along a traverse from Lhasa to Quxu in 1986 (Fig 1). The sampling localities are adjacent to the highway, between 16.5 km and 53 km from Lhasa (Duilongdeqing), and are located within 27 km of the Indus–Zangbo suture zone. Samples from two of the sites (T25 and T26) were from the D unit of Debon et al. [9] in the batholith, whereas the other seven sites were from the A and B units. A total of five samples of welded tuff were collected from the Lingzizong formation at site 31 northwest of Lhasa. This sampling site is at the same location as site 28 of Achache et al. [2].

Hand samples were taken and a magnetic compass was used for orientation. Each site typically comprised five independently oriented samples distributed over distance ranging up to 20 m. The present declination value at each sampling point was estimated from the calculation using the International Geomagnetic Reference of 1986 [19]

3. Paleomagnetism

3.1 Paleomagnetic measurements

Individual specimens (25 mm in diameter and 25 mm long) were prepared from each sample in the laboratory. Natural remanent magnetization (NRM) was measured with a ScT cryogenic magnetometer A specimen from each sample was treated in alternating magnetic fields (AF) up to 30 mT. The other specimens were subjected to stepwise thermal demagnetization up to 710°C.

Zijderveld component plots [20] and equal area projections were used to assess the directional stability and coercivity spectrum of each specimen. Principal component analysis [21] was used to estimate the direction presented by the colinear array of four or more data points defining the component of magnetization.

3.2 NRM

The intensity of NRM of samples of the Gangdese batholith varies only by a factor of 4, from 1.5×10^{-5} A/m to 4×10^{-6} A/m, except for sites T25 and T26 (see Fig. 2). Specimens from the latter two sites show intensity as low as 1×10^{-6} A/m. Site T25 and T26 belong to the D unit of Debon et al. [9], in which the plutonic rocks display a post-magmatic schistosity with partial recrystallization.

3 3. Demagnetization

Progressive thermal demagnetization separated samples into three categories (Fig. 2): (A) single

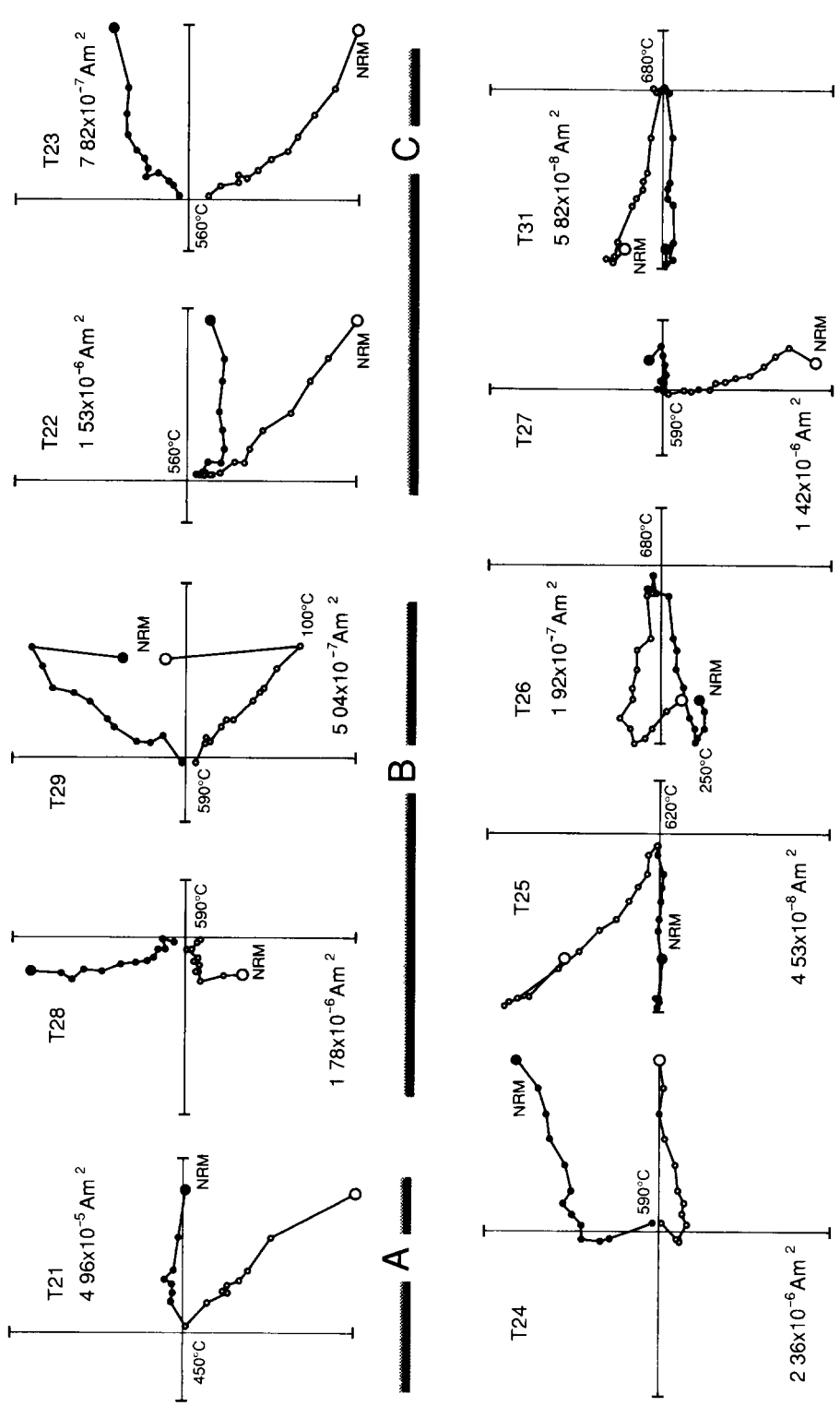


Fig 2. Orthogonal projections of magnetization vector end points during thermal demagnetization experiments for specimens in three categories, A, B and C (see text). Open symbols show the magnetic vector projection onto the vertical plane, solid symbols onto the horizontal plane Tectonic tilt corrections have not been applied

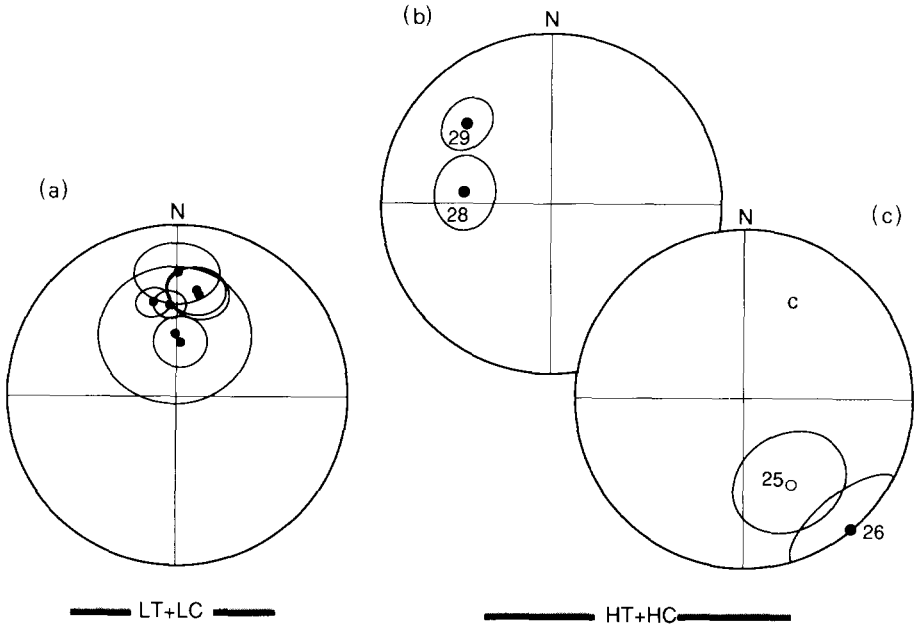


Fig 3 Equal area projection of mean site directions of three groups in the Quxu pluton (see text) (A) LT + LC component (T21 to T27) (B) HT + HC component of T28 and T29 (C) HT + HC component of T25 and T26 of the D unit of Debon et al [9]

component magnetization with low unblocking temperature, (B) single component magnetization with high unblocking temperature, and (C) two-component magnetization.

Category (A): Typical samples of this category are observed in site T21. Specimens lose their NRM down to 5% intensity of the original NRM after demagnetization to 450°C. The magnetiza-

TABLE 1

Paleomagnetic directions from the Quxu pluton and the Lingzizong formation All data, except for those of the Lingzizong formation, are without tilt correction Rock type GD = granodiorite, M = Mylonite, BG = biotite-bearing granite LT = low-temperature component, HT = high-temperature component, LC = low-coercivity component, HC = high-coercivity component $N(n)$ = number of samples and specimens A mean direction for each site is calculated by assigning unit weight to specimens D = declination, I = inclination, k = precision parameter, α_{95} = radius of 95% confidence Mean directions of T26 are calculated using ten specimens in five samples (*) and eight specimens in four samples (**) The Indus–Zangbo suture zone (ITS) is assumed to be located along 29°17 3′ at the Lhasa area [16]

Locality		Rock	Site	LT+LC					HT+HC					Distance
Lat	Lon	Type		N(n)	D(°)	I(°)	K	α_{95}(°)	N(n)	D(°)	I(°)	K	α_{95}(°)	from ITS(km)
Quxu Pluton														
29°21 5′	90°44′	GD	T21	5(10)	10 6	38 8	13 7	13 5						8
29°21 5′	90°45′	GD	T22	4(7)	-1 4	59 7	3 9	35 2						8
29°22′	90°47′	GD	T23	5(9)	-14 1	42 9	46 4	7 6						9
29°23′	90°50′	GD	T24	5(10)	0 2	27 2	8 8	17 2						11
29°23′	90°53′	M	T25	5(12)	-4 1	45 1	38 2	7 1	5(10)	151 8	-41 3	4 5	25 6	11
29°25′	90°55′	M	T26	4(8)	10 6	37 0	20 1	12 7	5(10)*	142 9	12 3	4 7	25 4	14
									4(8)**	140 6	0 4	6 3	24 1	
29°27′	90°56′	GD	T27	4(7)	4 4	64 0	23 7	12 7						18
29°30 5′	90°57′	GD	T28						5(11)	-82 8	45 4	8 6	16 5	24
29°32′	90°58′	BG	T29						5(10)	-47 6	31 0	21 6	12 2	27
Lingzizong Formation														
29°47′	90°47′	WT	T31(Before)	4(7)	-10 6	52 0	20 7	13 6	4(7)	-172 2	-17 4	34 2	10 5	55
			(After)	4(7)	15 4	61 3	20 7	13 6	4(7)	-165 4	-23 7	34 2	10 5	

tion trajectories are relatively straightforward, so the stable direction can be retrieved at the temperature range between room temperature and 300°C. The low unblocking temperature (LT) component is identical to the low-coercivity (LC) component: The LT component is found during AF demagnetization at low field between 0 mT and 10 mT.

Category (B): Thermal and AF demagnetization experiments on samples of T28 and T29 yield a univectorial magnetization. Unblocking is restricted to temperature around 580°C. This high-temperature (HT) component probably resides in titanium-poor magnetite. The HT component has higher coercivity (HC) than the LT component observed in category A; a similar direction of the HT component is identified up to 20 mT by AF demagnetization. Laboratory-induced magnetizations is erased by thermal demagnetization at 100°C.

Category (C): Granitic rocks from T22 to T27 and welded tuff of T31 have two-component magnetization. An LT component is destroyed at the temperature range between 200°C and 400°C. A HT component appears above 450°C. Demagnetization vectors reach the origin at around 580°C, with the exception of samples from T26. Trajectories of the latter samples converge toward the origin in the temperature range 630 to ~ 700°C. The HT component probably resides in either Ti-poor magnetite or haematite (T26). The LT component is observed during AF demagnetization at low field between 0 mT and 5 mT, while the HT component is identified above 10 mT. The HT component seems to be identical to the HC component, and the LT component to the LC component.

3.4 Characteristic directions

Mean directions of the LT + LC and the HT + HC components for each site were calculated by assigning unit weight to specimens (Fig. 3, Tables 1 and Table 2). We arbitrarily discarded results from any sites for which the circle of 95% confidence exceeded 36°. The HT + HC components for T22, T23, T24 and T27 failed this criterion.

The Quxu pluton: Three characteristic directions can be distinguished: (a) northerly direction of LT component magnetization, (b) westerly direction of HT component magnetization, and (c) southerly direction of HT component magnetization (Figs. 3a, 3b and 3c respectively).

TABLE 2

Average paleomagnetic direction of the Quxu pluton Direction is presented as the value at each representative point N = number of sites $\bar{D}$ = mean declination, $\bar{I}$ = mean inclination

	Sites	N	Reference position		$\bar{D}$(°)	$\bar{I}$(°)	k	α_{95}(°)
(A)	LT+LC	T21–T27	7	29°22'N	90°47'E	1 0	45 2	32 0	10 8
(B)	HT+HC	T28,T29	2	29°30 5'N	90°57'E	-63 3	39 3	--	--
(C)	HT+HC	T25,T26	2	29°23'N	90°53'E	145 3	-20 5	--	--

Characteristic directions of the LT component magnetizations are resolved at seven out of nine sites. Seven sites (T21 to T27) are located within 18 km of the suture zone. Their directions are northerly with normal polarity. The mean direction ($D = 1.0°$; $I = 45.2°$, $\alpha_{95} = 10.8°$) is parallel to the present axial dipole field ($D = 0°$, $I = 48°$).

The HT component directions are found at four sites. These are divided into two groups: The direction of the first (T28 and T29) is largely westerly with normal polarity. Site T28 and T29 have declinations of $-83°$ and $-48°$ respectively.

The other group, from the D unit (T25 and T26), shows reversed polarity. The positive inclination value of 12.3° for T26 can be reduced to 0 4° (see Table 1) if two data from one sample ($D = 147°$, $I = 71°$ and $D = 165°$, $I = 47°$) are ignored. The steep inclination values for this sample are probably due to overlap of the two component magnetizations of reversed and normal polarity. The site mean direction ($D = 140.6°$, $I = 0.4°$) from the remaining four samples is believed to best represent the characteristic direction of T26. The mean direction of T25 and T26 shows negative inclination and deflects anti clockwise from south by 35°.

The Lingzizong formation: The HT component from site T31 shows reversed polarity with a southerly direction, whereas the LT component reveals a northerly direction parallel to the present axial dipole field

4. Identification of magnetic minerals

The thermomagnetic analyses were carried out with a recording thermomagnetic balance on single specimens for each site at the batholith. These

specimens were heated in air in a magnetic field of 0.8T.

The thermomagnetic (J_s–T) curves for all specimens, except for those at sites T25 and T26, were quite similar to each other. As shown in Fig. 4, the specimens have single high Curie temperatures of around 580°C and reversible J_s–T curves. These curves suggest that the major carrier of the magnetization is Ti-poor magnetite. Microscopic observations support this interpretation: the most common opaque mineral is magnetite subdivided by ilmenite lamella which are partly altered to hematite.

Thermomagnetic curves of T25 and T26 contrast with the others (Fig. 4). Specimen of T26 shows an asymptotic decrease in saturation magnetization with increasing temperature. This behavior links a predominance of paramagnetic carriers obeying the Curie Law. Sample T25 exhibits an irreversible two-phase J_s–T curve. The lower Curie temperature in the two phases is around 300°C, while the higher one is around 580°C. The two phase behavior might be characteristic of the coexistence of unoxidized Ti-poor magnetite and oxidized Ti-poor magnetite which had undergone low-temperature oxidation to maghemite. Micro-

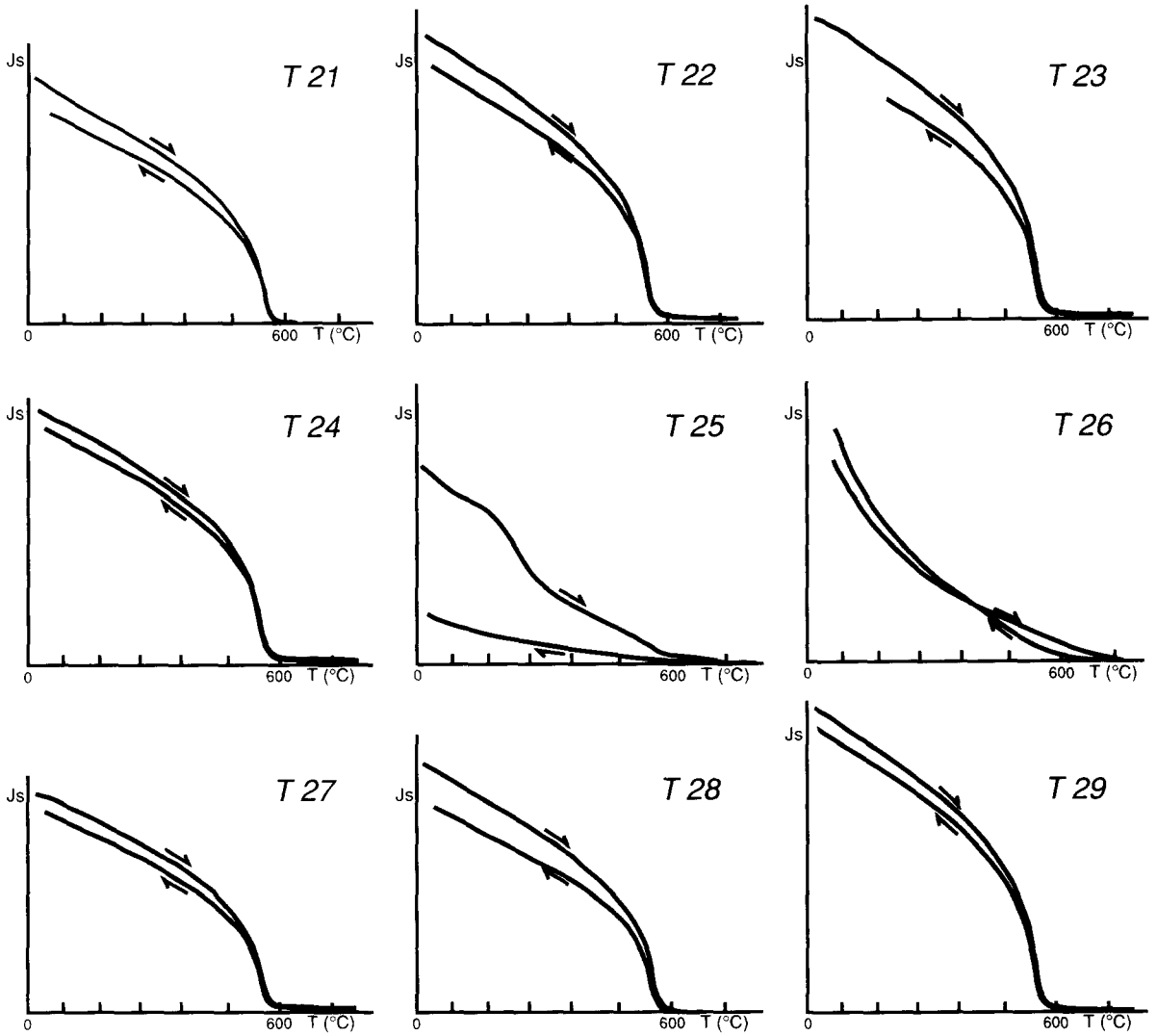


Fig 4 High field saturation magnetization versus temperature (J_s–T) curves for the Quxu pluton samples (T21 to T29) Experiments were performed in air Field intensity was 0 8 T The arrows indicate heating and cooling

scopic observation reveals that the predominant magnetic mineral in both specimens T25 and T26 is, however, hematite. No magnetite or maghemite was identified.

5. Discussion

5.1 Counter-clockwise rotation of the Quxu pluton

Paleomagnetic and rock magnetic studies suggest that the westerly direction of the HT component of T28 and T29 in the Quxu pluton is of primary origin. Thermal demagnetization and thermomagnetic analysis reveal that their magnetization is carried by magnetite (Figs. 2 and Fig. 4). The microscopic observations show that the granodiorite (T28) and biotite-bearing granite (T29) are quite fresh plutonic rocks in which magnetite is the predominant magnetic mineral. This magnetite is probably original material, as it shows a euhedral shape. There is no evidence that the batholith has experienced deformation or metamorphism, suggesting that the magnetite has been subjected to little chemical change. These observations lead us to the conclusion that the magnetic carriers of magnetite have retained their primary magnetization.

The westerly directions of the primary component have declinations of $-48°$ and $-83°$. These directions contrast with those of the Cretaceous rocks to the north of the batholith (Fig. 5). They have a northerly direction [2,22]: the Takena formation shows a declination of $-6°$ to ca. $-3°$ and the Lingzizong formation shows values of $-9°$ to $+14°$. The Quxu pluton has a direction that is distinct from those of the other formations in its vicinity.

The mean direction of the Quxu pluton ($D = -63°$) is quite consistent with a largely counter-clockwise deflection in the declination of the Cretaceous and Paleogene rocks along the Indus–Zangbo suture zone [7,8] (Fig. 6). The declination values of the latter rocks range from $-85°$ to $-61°$. Their inclination values (31° and 45°) also fall within the range of those observed along the suture zone ($-18°$ to $+52°$) [7,8]. This good agreement in paleomagnetic directions suggests that the Quxu pluton was subjected to the same paleomagnetic rotation with a counter-clockwise sense as the blocks along the suture zone.

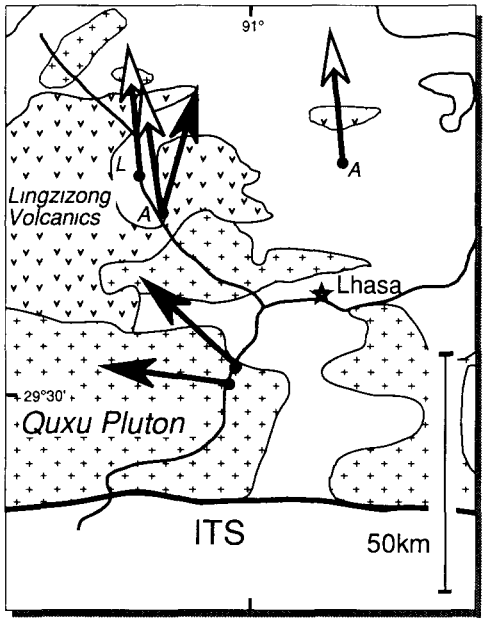


Fig 5 Paleomagnetic directions of the Quxu pluton and Lingzizong formation, together with published data for the Lingzizong and Takena formations The solid arrows were obtained from this study, the open arrows from previous studies (A from [2] and L from [22]) *ITS* = Indus–Zangbo suture zone

5.2 Deformation of the narrow zone along the suture zone

The Quxu pluton provides information on the areal extent and timing of deformation along the suture. The Quxu pluton is the easternmost site where the largely westerly paleomagnetic rotation is observed: the sampling localities T28 and T29 are at 90°57′E and 90°58′E respectively (Fig. 6) The locality of the westernmost site with the largely counter-clockwise rotation of 65° is Jiangba, at 82°20′E [8]. The crustal blocks with the counter-clockwise rotation are distributed between 82°E and 91°E. The length of the deformed zone extends to 840 km along the suture.

Deformation is distributed over a ca. 30 km region within the Asian continent: sampling localities T28 and T29 with the westerly direction are 24 km and 27 km from the Indus–Zangbo suture zone (Fig. 5). Deformation does not, however, propagate to the region around 55 km from the suture, because little paleomagnetic rotation is recorded in the Takena formation and Lingzizong formation. The northern boundary of the de-

formed zone lies between 27 km and 55 km from the suture zone.

Sampling sites T28 and T29 are close to the localities where $^{40}Ar/^{39}Ar$ and U-Pb ages of 42.5 Ma were obtained [4,10]. This indicates that the counter-clockwise rotation of the Quxu pluton occurred after 42.5 Ma. On the basis of a number of independent methods [3-5,10] the collision of India with Asia is known to have occurred during Late Eocene time, around 45 Ma. The timing of the counter-clockwise rotation of the Quxu pluton implies that the deformation of the crustal blocks along the suture zone may be ascribed to collision and subsequent indentation of India.

Although, from laboratory and numerical experiments [23–28], various deformation modes have been expected, the "domino style" deformation model [8] is the plausible explanation for a constant amount of rotation in the long, narrow area along the suture zone (see Fig. 6). The narrow zone, which is 840 km long and less than 100 km wide, successfully absorbed tangential stress under the conditions of a compressive regime brought about by the collision of India during the process of the domino style deformation.

There is little evidence of the domino style deformation in the topography. Any topographic features produced during counter-clockwise rotation are probably masked by recent tectonism After a long period of north–south shortening since the collision of India, southern Tibet is now undergoing east–west extension [29–36]. The resulting extensional regime in southern Tibet led to development in the Pliocene of north–south trend-

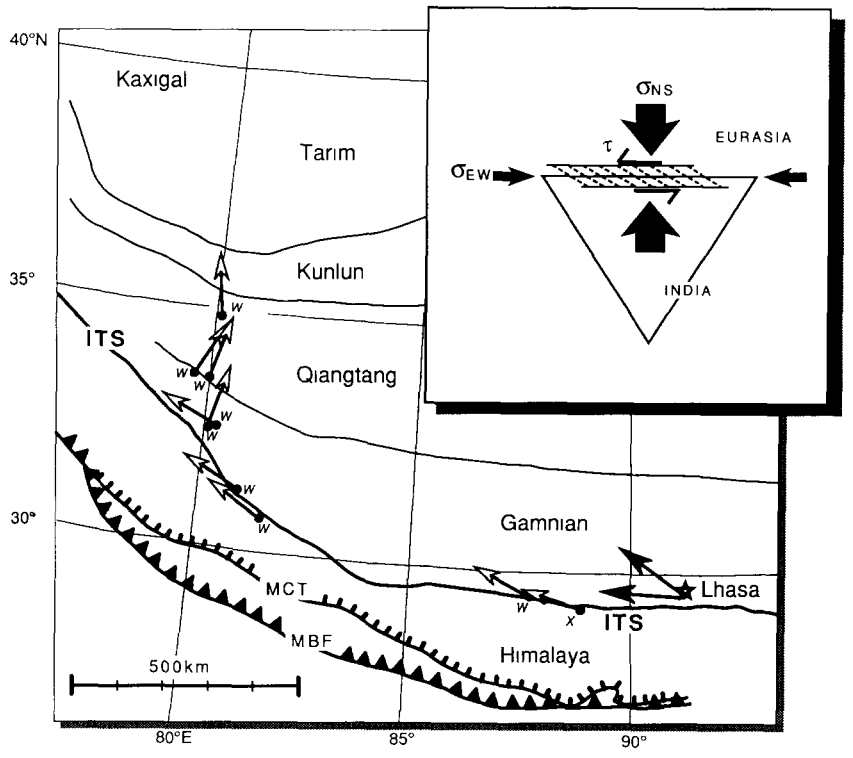


Fig 6 42.5 Ma paleomagnetic directions of the Quxu pluton with the published Cretaceous and Paleogene paleomagnetic directions near the Indus–Zangbo suture zone The solid arrows were obtained from this study and the open arrows from the previous studies in western Tibet (w) [8] and Xigaze (x) [7] The major terranes in Tibet are shown Heavy line is the Indus–Zangbo suture zone (ITS), and barbed lines are continental thrusts (MCT = Main Central Thrust, MBF = Main Boundary Fault). The inset shows the "domino style" deformation model proposed by Otofuji et al [8] As the north-south normal stress (σ_{NS}) increased up to the critical state, the crustal blocks began to slide along the slip planes (dotted lines) The left-lateral shear (τ) set in the "domino style" deformation with lateral displacement to the left in the narrow zone between the Indian and Eurasian continents

ing normal faults [32]. These presumably hide the geomorphological evidence of the deformation of the collision period, around 42.5 Ma.

5 3 Secondary magnetization component in the Quxu batholith

HT component with southerly direction: The batholith of the D unit (T25 and T26) has HT component magnetizations with the southerly direction (Fig. 3). The magnetization of the D unit has a character that is quite distinct from that of the other sites· the weak NRM is carried by hematite as well as magnetite. Hematite predominates in microscopic observations.

The HT component of the D unit may be ascribed to secondary magnetization which originated during cataclastic metamorphism. In the microscopic observations, the most conspicuous structure is porphyroclastic texture: the margins of phenocrysts are destroyed, and finely recrystallized materials (e.g., sericite, biotite and smectite) occupy the interstices between phenocrysts Deformation lamellae are observed in plagioclase These observations strongly indicate that the mylonitization due to cataclastic metamorphism occurred in the D unit under conditions of strong shear stress. Most metamorphic rocks are likely to acquire complicated magnetizations [37]. New, but weak CRM was probably generated in hematite which formed by oxidation of magnetite during the cataclastic metamorphism. Unoxidized magnetite grains were presumably subjected to pressure effects (pressure demagnetization of their original magnetizations and rotation of their magnetic vectors) [38–41].

The HT secondary component of the D unit reveals counter-clockwise paleomagnetic rotation of about 35°. The amount of this rotation is smaller than that of the primary magnetization direction of about 60° recorded in T28 and T29, suggesting that the secondary component in the D unit was produced at sometime during counter-clockwise rotation in the domino style deformation process. The D unit may have behaved as a slip plane during the domino style deformation (see Fig. 6), in which strong shear stress generated the cataclastic metamorphism.

LT component with northerly direction: The LT component is ascribed to thermoviscous remanent magnetization [42–44]. Distinct LT component directions are northerly with normal polarity and their mean direction is parallel to the present axial dipole field, within the 95% confidence limit (Fig. 3). This component was easily erased by both AF demagnetization up to 10 mT and thermal demagnetization in the 200–350°C temperature range (Fig. 2). Unblocking temperatures of 200°C and 350°C are expected to be acquired at room temperature over a scale of 1 m.y. for magnetite and hematite, respectively [42,43]. Magnetite and hematite at the site of T21 to T27 could have acquired secondary thermoviscous remanent magnetization in the recent normal polarity epoch.

Few distinct HT components appear in the samples from T21 to T24 close to the suture zone (Fig. 1). The HT component of the original remanent magnetization was probably destroyed. The destruction of the HT component is not, however, attributed to metamorphic processes, because on the basis of microscopic observation these rocks were subjected to little deformation or metamorphism. An alternative explanation for the destruction is demagnetization during slow cooling by subsequently frequent geomagnetic field reversals [45]. The slow cooling is probably due to slow uplift of the batholith and high heat flow in southern Tibet. Fairly high heat flow of 146 mW m^{-2} is reported over a distance of 30 km south of the Indus–Zangbo suture zone [46,47]. Uplift rates for the Gangdese belt vary from 0.07 mm/yr to 4 mm/yr in both space and time [10]: Rapid uplift is expected for the northern part of the Quxu batholith at site T28 and T29 where the HT component is preserved [10]. The southern part of the Quxu batholith close to the suture zone may rise at a fairly slow rate. After the original magnetization was destroyed during slow cooling, only thermoviscous remanent magnetization remained

6. Conclusion

(1) The Quxu pluton at 90°58′E, 27 km from the Indus–Zangbo suture, was subjected to counter-clockwise rotation of about 60°. The rotation occurred after 42.5 Ma.

(2) No counter-clockwise rotation is recorded in the Cretaceous welded tuffs of the Lingzizong formation 55 km from the suture.

(3) The Quxu pluton was rotated in the "domino style" deformation process as a part of the de-

formed zone along the Indus–Zangbo suture after 42.5 Ma. The deformed zone is more than 840 km long and less than 100 km wide.

(4) The batholith of the D unit in the Quxu pluton acquired secondary magnetization during cataclastic metamorphic processes at sometime during the domino style deformation

(5) The batholith within 11 km the Indus–Zangbo suture zone only records the thermoviscous remanent magnetization parallel to the present axial dipole field. The primary remanent magnetization may have been destroyed during slow cooling at the uplift stage.

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